

What Drives SENSORS



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A sensor-like accelerometer is not something new, but it has been miniaturised and volume production has made it so cost-effective that you can put it almost everywhere. “Sensors started as a component in numerous applications such as mobiles and tablets, but you can now see these in industrial equipment, toys, medical electronics, wearables and even safety in vehicles,” says Vivek Sharma, regional vice president, greater China and South Asia region - India operations, director - India design centres, STMicroelectronics, in an interview with EFY.

What is driving these little pieces of technology that are so crucial for the Internet of Things (IoT)? Could sensors be the new linen?

The IoT has had a great influence on sensor technology. Whether it is wearables, implantables, smartfabrics or smartpills, improvements in micro-electro-mechanical systems (MEMS) technology and sensors have been one of the driving factors.

Hybrid circuits

Walden C. Rhines, chairman and CEO of Mentor Graphics, told us in an interview that the IoT has created a greater interest in multi-die packaging, in analogue and RF for relatively-low-complexity designs and in hybrid circuits that combine sensors, actuators, MEMS and other

Bird's eye view of driving elements

- Hybrid sectors
- Compression happening in MEMS
- NEMS
- Sensor patches and tattoos
- Flexible image sensors
- Large-area integration of micro-electronics
- Integrated smartsensors

things with ASIC chips and then package these accordingly.

Firms like PragmatIC Printing are developing ultra-thin, low-cost flexible microcircuits that can be incorporated in mass-market packaging. Steven Bagshaw, marketing executive at Centre for Process Innovation Ltd also mentioned how hybrid electronics will give rise to wireless medical devices for rapid diagnostics using printed sensors, thus helping build better medical devices in line with the IoT concept.

Centre for Nano Science and Engineering (CeNSE) at Indian Institute of Science (IISc) is also working on nano-electro-mechanical systems (NEMS). Dr Vijay Mishra, CTO at CeNSE in IISc, has presented a talk on nanotech sensors for human body health monitoring at the 4th edition of Electronics Rocks conference that took place in Bengaluru last year.

Scaling up sensor deployment

John Rogers' materials science research team from University of Illinois, USA, has developed a way of building circuits that act like tattoos, collect power wirelessly and can be worn just about anywhere on the body. Their sensors harvest energy through near field communication to power themselves. Rogers' company named MC10 Inc. has marketed these in the form of sensor patches, and they are now working on a new generation of the technology since late 2014.

Technical drawing of the latest version of Sensor Fish

